Fixed-Multiplicity Tree Fibres and Five-Point Liftings for the Five-Cycle Double Cover Conjecture

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Resolution status. The standard five-cycle double cover conjecture remains unresolved. This paper proves reductions and exact criteria inside one proof route, gives certified counterexamples to several tempting stronger lemmas, and reports bounded positive computations. It proves neither the conjecture nor its negation.

Abstract

The five-cycle double cover conjecture asks whether every finite bridgeless graph has at most five Eulerian edge-subsets covering every edge exactly twice. We study a fixed-multiplicity spanning-tree route suggested by Jaeger's nowhere-zero $\mathbb{F}_2^3$-flow construction.

For a cubic graph and three spanning trees with empty common intersection, the edge multiplicities have only two possible defect types. We prove that every prescribed defect pattern of either type is feasible in every 3-edge-connected cubic graph, by a direct Nash-Williams–Tutte partition argument. Blasiak's graphic-matroid theorem then identifies each fixed-multiplicity fibre with one symmetric-exchange component.

We give a self-contained derivation of a component-parity criterion. Given a nowhere-zero $\mathbb{F}_2^3$-flow and a prescribed five-point subset of the eight affine points, a compatible pair labelling exists exactly when one forced leaf-bit xor vanishes in every component of a distinguished value subgraph. Hušek and Šámal independently obtained the same flow-level characterization in Theorem 3.16 of [6]; we claim no novelty or priority for this criterion. For flows built from spanning trees, it becomes an odd-kernel statement: if $K(T)$ is the unique all-vertex-odd forest contained in T, then the star-fibre target is that $K(T_1) \cap K(T_2)$ be Eulerian after contracting every component of $K(T_3)$.

We prove the exact tree-exchange law for these odd kernels. We also give small, directly checkable obstructions to three natural strengthenings: Petersen forbids pairwise-disjoint odd kernels; the cube refutes a fixed-third-tree repair lemma; and a ten-vertex Möbius ladder refutes an unqualified "choose perfect forests first, extend later" strategy. A 34-vertex strong snark, represented by a 4,444-variable CNF and a dual-checked LRAT certificate, refutes every flow-direction thinning rule in one star fibre, while the same fibre has an independently checked five-point labelling.

We also represent $K(T) \cup \{p\}$ as the fundamental circuit of one all-ones parity element p in a binary extension of the graphic matroid. Deleting the root of a star fibre turns its three-tree packings into partitions into three cographic bases with three prescribed terminal joins. The sufficient strengthening $K(T_1) \cap K(T_2) \subseteq \mathrm{cl}(K(T_3))$ is false: an exact 12-vertex enumeration refutes it for generic type-A fibres, and a dual-checked 16-vertex certificate refutes it even for a vertex-star fibre. Both countermodels preserve the weaker, correct quotient-parity target. A human triangle-contraction theorem extends the latter closure failure to an infinite rooted family, without making any claim against FiveCDC. In contrast, the exact component-parity target is invariant in both directions under a nonroot triangle expansion; the lifting proof reduces to seven human-checkable local trace rows.

Exact computation finds a five-point representative in all 804,204 feasible type-A/type-B fibres of all 587 connected simple bridgeless cubic graphs through order 14; in all 26,790 vertex-star fibres of 705 order-38 strong snarks; and in all 1,364 vertex-star fibres of a retained family of 31 order-44 oddness-four snarks. Only the last census retains and independently replays every literal witness. A separate exhaustive exchange census proves, through order 14, that every positive same-level component for the minimum seven-plane parity defect has an edge to a lower level; it covers 529,150,122 exact states. We derive the exact simultaneous seven-plane update under a reciprocal tree exchange. Two explicit 16-vertex states show that neither immediate descent nor descent exposure inside one fixed odd-kernel triple is valid; active same-level motion is sometimes necessary. A 36-vertex state further shows that even the complete sorted seven-defect profile need not decrease in one exchange, although an exact two-exchange same-level escape exists. For square expansion, we prove that a fixed five-cover extends exactly when the two deleted edge labels are equal or disjoint, and a solver-free whole-fibre census finds compatible state-and-lift choices in every simple 3-edge-connected cubic instance through downstairs order 14. These are finite results and exact no-go statements, not a universal selection theorem.

Contents

1 Status, conventions, and contribution boundary

Oum defines a k-cycle double cover as a collection of at most k Eulerian subgraphs covering every edge exactly twice, proves the eight-subgraph bound, and states the five-subgraph assertion as Conjecture 18 [7]. The conjecture is attributed to Celmins and Preissmann [3, 14]; see also the surveys and flow work of Jaeger [8, 9, 10] and recent special cases [11, 5].

Hušek and Šámal's contemporaneous preprint, submitted July 27, 2026, explicitly records FiveCDC as open (their Conjecture 1.9), proves an exact five-support component-parity characterization (their Theorem 3.16), and states the resulting equivalent flow-selection problem as Conjecture 3.19 [6]. Their theorem independently establishes the flow-level content of Theorem 4.2 below. The missing issue in both formulations is the universal choice of a suitable flow.

Throughout, a graph is finite and undirected. Parallel edges are distinct edge objects. Unless a statement says "simple", parallel edges are permitted. A loop contributes two incidences at its endpoint. An *Eulerian edge-subset* is an edge set whose incidence degree is even at every vertex; it need not be connected or 2-regular.

The five members form an indexed tuple. Empty Eulerian subsets and repeated subsets are allowed. Hence "five, allowing empty members" is equivalent to "at most five". This is the standard convention used in [7]. The orientable five-cycle double cover conjecture imposes directed Eulerian orientations that use every edge once in each direction. It is a distinct, stronger conjecture and is not considered here.

We explicitly withdraw any novelty or priority claim for the component criterion: [6, Theorem 3.16] independently gives the equivalent characterization. The specialization to Jaeger star fibres, odd kernels, triangle invariance, and the exchange and no-go results may be useful independently, but they still require a specialist prior-art review. The paper is a research draft whose proofs and certificates are deliberately exposed for human audit.

2 The exact cover encoding

Let $G = (V, E)$ be an edge-indexed multigraph. For $e \in E$ and $i \in \{0, 1, 2, 3, 4\}$, introduce $x_{e,i} \in \{0, 1\}$. Write $\mathrm{inc}(v)$ for the multiset of incidences at v; a loop appears twice. Consider

$$\sum_{i=0}^{4} x_{e,i} = 2 \qquad (e \in E), \tag{2.1}$$

$$\bigoplus_{(e,\ \text{incidence})\in \mathrm{inc}(v)} x_{e,i} = 0 \qquad (v \in V,\ 0 \le i < 5). \tag{2.2}$$

Theorem 2.1 (encoding equivalence)**.** *Satisfying assignments of* (2.1)–(2.2) *are in bijection with ordered five-tuples* $(C_0, \dots, C_4)$ *of Eulerian edge-subsets that contain every edge in exactly two coordinates.*

Proof. Given an assignment, set $C_i = \{e : x_{e,i} = 1\}$. Equation (2.1) says that each edge occurs in exactly two of these sets. Equation (2.2) is exactly the degree of v in C_i, reduced modulo two. Hence every C_i is Eulerian.

Conversely, take the five characteristic functions of a cover. Exact double coverage gives (2.1), and Eulerian parity gives (2.2). The two maps are inverse. A loop causes no exception: its two parity incidences cancel, while its single membership variable remains constrained by (2.1). $\square$

This gives both a native-XOR encoding and a certificate-producing CNF. For exact-two among $y_0, \ldots, y_4$, use the ten negative clauses on triples and the five positive clauses on four-subsets. For an even-parity row $z_1 \oplus \cdots \oplus z_d = 0$, add the blocking clause

$$\bigvee_{a_j=1} \neg z_j \;\vee \bigvee_{a_j=0} z_j$$

for every odd vector $a \in \{0,1\}^d$. This direct form has no parity auxiliaries. It is exponential in d, but small on the cubic search domain. An UNSAT proof for this formula certifies the fixed-graph nonexistence claim only after graph parsing, incidence conventions, and bridgelessness are checked separately.

3 Tree coordinates and fixed-multiplicity fibres

3.1 Fundamental completions

Let T be a spanning tree of a connected graph G. For every cotree edge e, let $C_T(e)$ be its fundamental circuit, and define

$$F(T) = \mathop{\triangle}_{e \notin T} C_T(e). \tag{3.1}$$

Lemma 3.1 (unique completion)**.** *$F(T)$ is the unique Eulerian edge-subset containing every edge of $E(G) - T$.*

Proof. Every fundamental circuit is Eulerian, so their symmetric difference is Eulerian. Each cotree edge belongs to its own fundamental circuit and to no other one, hence belongs to $F(T)$. If F' is a second Eulerian set containing the cotree with the same prescribed cotree indicators, then $F(T) \triangle F'$ is an Eulerian subset of the tree T. A nonempty forest has a vertex of odd degree in its selected subgraph, so that difference is empty. $\square$

Let T_1, T_2, T_3 be spanning trees with empty common intersection. Put $F_i = F(T_i)$ and

$$\phi(e) = \big(\mathbf{1}_{F_1}(e), \mathbf{1}_{F_2}(e), \mathbf{1}_{F_3}(e)\big) \in \mathbb{F}_2^3. \tag{3.2}$$

Each coordinate is Eulerian, so ϕ is a flow. Every edge is omitted by some T_i, hence is in F_i, and therefore $\phi(e) \neq 0$. This is Jaeger's spanning-tree construction of a nowhere-zero $\mathbb{F}_2^3$-flow; compare [8, 7].

Definition 3.2 (compatible pair labelling)**.** For a nowhere-zero flow $\phi : E(G) \to W - \{0\}$, where $W = \mathbb{F}_2^3$, a compatible pair labelling is a map

$$\lambda(e) = \{x_e, y_e\} \in \binom{W}{2}$$

such that $x_e + y_e = \phi(e)$, and such that every point $s \in W$ occurs an even number of times among the pairs incident with every vertex.

Proposition 3.3 (pair labels give covers)**.** *If a compatible pair labelling uses at most five points of W, then its point-coordinate edge sets are a five-cycle double cover.*

Proof. For each used point s, take $H_s = \{e : s \in \lambda(e)\}$. Compatibility makes H_s Eulerian. Every pair has two distinct points, so every edge lies in exactly two H_s. Pad with empty sets if fewer than five points are used. $\square$

3.2 Three defects

Now suppose G is cubic on n vertices. Define the tree multiplicity

$$m_e = \big|\{i : e \in T_i\}\big|, \qquad d_e = 2 - m_e. \tag{3.3}$$

Since the common intersection is empty, $m_e \leq 2$, and

$$\sum_{e \in E(G)} d_e = 2\frac{3n}{2} - 3(n-1) = 3. \tag{3.4}$$

Thus there are exactly two patterns:

1. *type A*: three distinct edges have $d_e = 1$;
2. *type B*: one edge has $d_e = 2$, a different edge has $d_e = 1$.

All other defects vanish.

Theorem 3.4 (prescribed-fibre feasibility)**.** *Let G be a finite 3-edge-connected cubic graph. Every vector $d \in \{0,1,2\}^{E(G)}$ with $\sum_e d_e = 3$ is realized by three spanning trees whose multiplicity on e is $2 - d_e$.*

Proof. Replace each edge e by $2 - d_e$ labelled parallel copies, obtaining a multigraph H_d. It has

$$|E(H_d)| = 2|E(G)| - 3 = 3(n-1). \tag{3.5}$$

Consider a partition $\mathcal{P}$ of $V(G)$ into $k \geq 2$ nonempty parts. Let q be the number of original edges crossing the partition and let D be the sum of d_e over crossing edges. Every part is a proper nonempty vertex set, so 3-edge-connectivity gives boundary size at least three. Summing the k boundary sizes counts each crossing edge twice; hence $2q \geq 3k$. Since $D \leq 3$,

$$|E_{H_d}(\mathcal{P})| = 2q - D \geq 3k - 3 = 3(k-1). \tag{3.6}$$

By the Nash-Williams–Tutte spanning-tree packing theorem [12, 17], H_d has three edge-disjoint spanning trees. Their total size is $3(n-1) = |E(H_d)|$, so they use every copy. Projecting labelled copies back to G gives the prescribed multiplicities. $\square$

Thus every type-A and type-B placement is feasible in the 3-edge-connected cubic domain; the fibre is never empty there.

3.3 Exchange connectivity

A reciprocal symmetric exchange replaces two trees by

$$T_i' = T_i - e + f, \qquad T_j' = T_j - f + e, \tag{3.7}$$

when both displayed sets are spanning trees. It preserves the edgewise multiplicity vector and the pairwise intersection $T_i \cap T_j$, hence also preserves an empty triple intersection.

Blasiak proved that the toric ideal of a graphic matroid is generated by quadrics corresponding to single-element symmetric exchanges [1]. Under the standard Markov-basis interpretation, any two multisets of graphic-matroid bases with the same multiset union are connected by these moves. Consequently:

Proposition 3.5 (fixed-fibre connectivity)**.** *After quotienting by permutation of the three trees, the symmetric-exchange components are exactly the feasible fixed-m fibres.*

The universal proof obligation inside this route is therefore not connectivity or feasibility. It is the following selection assertion.

Open problem 3.6 (fixed-fibre five-point selection)**.** Does every type-A or type-B fibre of every 3-edge-connected cubic graph contain a tree triple whose completion flow admits a compatible pair labelling supported on at most five points of $\mathbb{F}_2^3$?

This problem is stronger than what is needed for a single graph, since one good fibre would suffice for that graph. It remains open.

4 A five-point component-parity criterion

This section gives an exact, noncomputational test for the conclusion of Problem 3.6 once the flow and the five-point set are fixed.

Let $W = \mathbb{F}_2^3$, let $\phi : E(G) \to W - \{0\}$ be a nowhere-zero flow on a loopless cubic graph, and let $B \subset W$ have three points. The affine span of B is a four-point plane

$$A = b + U_0,$$

where $U_0 \leq W$ has dimension two. Write

$$a^* = A - B, \qquad A' = W - A, \qquad S = W - B = A' \cup \{a^*\}. \tag{4.1}$$

Choose $z \in A'$. Let $h_0 \in W^* - \{0\}$ have kernel U_0.

At a graph vertex v, the three incident flow values are distinct, nonzero, and sum to zero. Together with zero they form a linear plane U_v. Let h_v be its nonzero normal, and put

$$E_0 = \{e : \phi(e) \in U_0\}. \tag{4.2}$$

Lemma 4.1 (local triangle classification)**.** *In a compatible pair labelling, the three incident pairs at a cubic vertex are the three edges of a simple triangle on W. Their differences are the three nonzero points of U_v.*

Moreover, the degree of v in (V, E_0) is three if $U_v = U_0$, and one otherwise. A degree-three vertex has exactly four local triangles supported in S, while a degree-one vertex has exactly one.

Proof. View the three pair labels as the edges of a loopless three-edge multigraph on vertex set W. Local compatibility says that every vertex of this multigraph has even degree. The only nonempty loopless three-edge possibility is a simple triangle. If its vertices are r, s, t, its edge differences are

$$r + s, \quad r + t, \quad s + t,$$

which are the three nonzero elements of a two-dimensional linear subspace. They are exactly the three incident flow values.

The planes U_v and U_0 either agree or meet in a one-dimensional subspace. This proves the degree assertion. If they agree, a local triangle is three points of one coset of U_0. It cannot lie in A, because $A \cap S = \{a^*\}$; it is therefore $A' - \{t\}$ for one of the four $t \in A'$.

In the remaining case write the incident values as p, q, r, where $p \in U_0 - \{0\}$, $q, r \notin U_0$, and $p = q + r$. Every coset of U_v meets A in one p-pair and A' in the other. The coset containing a^* meets A in $\{a^*, a^* + p\}$. A supported three-point subset of that coset must omit $a^* + p$, and is therefore

$$\{a^*, a^* + q, a^* + r\}. \tag{4.3}$$

The other coset meets A in a p-pair contained entirely in B; omitting only one point cannot avoid B. This proves uniqueness. □

Call the degree-three vertices of (V, E_0) *special*, and its degree-one vertices *leaves*. For $p \in U_0 - \{0\}$, the affine plane A' has two p-pairs. Give $\{z, z + p\}$ port bit 1 and the other pair port bit 0.

At a special vertex with incident E_0-values p_1, p_2, p_3, the four triangles $A' - \{t\}$ produce exactly

$$000, \quad 011, \quad 101, \quad 110. \tag{4.4}$$

Thus their port bits obey one even-parity equation. At a leaf v, the unique local triangle forces the port bit on its only E_0-edge; call it ℓ_v. If $w = z + a^*$, the forced value has the coordinate-free description

$$\ell_v = (h_v + h_0)(w). \tag{4.5}$$

Indeed, the forced p-pair is $\{a^* + q, a^* + q + p\}$. It equals the base pair $\{z, z + p\}$ exactly when $w + q \in \{0, p\}$, equivalently when $w \in U_v$. Since $w \notin U_0$, we have $h_0(w) = 1$; hence the indicator of $h_v(w) = 0$ is precisely $(h_v + h_0)(w)$. For edges outside E_0, the unique labels forced by their two endpoints agree automatically.

Theorem 4.2 (component-parity criterion; cf. Hušek–Šámal)**.** *A compatible pair labelling supported in $S = W - B$ exists if and only if for every connected component Q of $(V(G), E_0)$,*

$$\bigoplus_{v \in L(Q)} \ell_v = 0, \tag{4.6}$$

where $L(Q)$ is the set of degree-one vertices of Q.

Proof. Give every edge $e \in E_0$ one variable $x_e \in \mathbb{F}_2$, its selected port bit. Lemma 4.1 gives exactly the equations

$$x_e = \ell_v \quad \text{at a leaf } v, \qquad \bigoplus_{e \ni v} x_e = 0 \quad \text{at a special vertex } v. \tag{4.7}$$

Equality of endpoint bits is precisely compatibility on edges of E_0; compatibility off E_0 is automatic by the unique local classification.

On a connected component Q, system (4.7) is its vertex-edge incidence system over $\mathbb{F}_2$, with right-hand side ℓ_v at leaves and zero at special vertices. The image of the incidence matrix of a connected loopless graph is exactly the even-parity subspace. Therefore the system is soluble exactly when the xor of the right-hand side is zero, which is (4.6). A solution selects one of the four permitted triangles at every special vertex and reconstructs the complete pair labelling. □

Remark 4.3 (priority and coordinate translation). Hušek and Šámal independently derive the same characterization in different coordinates [6, Theorem 3.16]. In their normalization the five allowed affine points are $\{000, 001, 010, 011, 100\}$; their eliminated local system is soluble exactly when every component of $(V, E - F)$ contains an even number of vertices from their set Z. After an affine relabelling taking S to that five-set, $E - F$ is our distinguished value subgraph E_0, and their component right-hand side is the leaf xor in (4.6). Thus the proof above is a self-contained coordinate derivation, not a novelty claim. Their Conjecture 3.19 is the equivalent remaining task of choosing a nowhere-zero $\mathbb{F}_2^3$-flow for which this parity condition holds.

Remark 4.4. Cycles cause no exception. If Q has cycle rank $\beta(Q)$, then a soluble component has $2^{\beta(Q)}$ port solutions. The condition does not depend on the auxiliary base point z; changing z performs a gauge change on the edge variables.

An independent checker exhausts the local classification and all 30,744 literal nowhere-zero $\mathbb{F}_2^3$-flows on the eight connected simple cubic graphs through order eight. Direct solution of the original supported-triangle constraint and criterion (4.6) agree in every case. Not every arbitrary flow passes: 2,520 of these flows have no supporting plane at all.

5 Odd kernels and the star-fibre target

For a spanning tree T in a cubic graph of even order, define

$$K(T) = \{e \in T : \text{the components of } T - e \text{ have odd order}\}. \tag{5.1}$$

Theorem 5.1 (odd-kernel identity)**.** *$K(T)$ is the unique subset of T having odd degree at every vertex, and*

$$F(T) = E(G) - K(T). \tag{5.2}$$

In particular, $K(T)$ is a spanning all-vertex-odd forest and each of its components has even order.

Proof. Root T. For a nonroot vertex v, sum the required odd degrees over the subtree below v. Internal edges count twice, so the parent edge must be selected exactly when the subtree has odd order. This forces all nonroot parent-edge choices. The root equation follows because the total order is even. Thus the all-odd subset exists, is unique, and is exactly (5.1).

The complement $E(G) - K(T)$ is Eulerian because G has degree three and $K(T)$ has odd degree at every vertex. It contains every cotree edge. Lemma 3.1 gives (5.2). Summing odd degrees in one component of $K(T)$ shows that its number of vertices is even. □

Fix a vertex-star type-A fibre and number the spokes so that spoke i belongs only to T_i. Put $K_i = K(T_i)$. Under $\phi = (\mathbf{1}_{F_1}, \mathbf{1}_{F_2}, \mathbf{1}_{F_3})$, the zero set of coordinate three is K_3. Since the three trees have empty common intersection, $K_1 \cap K_2 \cap K_3 = \varnothing$, and the pure third-coordinate value class is exactly

$$\{e : \phi(e) = (0, 0, 1)\} = K_1 \cap K_2. \tag{5.3}$$

Taking $U_0 = \ker(x_3)$ in Theorem 4.2 yields:

Corollary 5.2 (star-fibre quotient criterion). *The canonical five-point set associated with* $\ker(x_3)$ *supports a compatible pair labelling if and only if*

$$|\delta_G(Q) \cap K_1 \cap K_2| \equiv 0 \pmod 2 \tag{5.4}$$

for every component Q *of* K_3*. Equivalently,* $K_1 \cap K_2$ *is Eulerian after every* K_3*-component is contracted.*

Some parity is automatic but insufficient. For every K_3-component Q, Theorem 5.1 gives

$$|\delta(Q) \cap K_1| \equiv |\delta(Q) \cap K_2| \equiv |Q| \equiv 0 \pmod 2. \tag{5.5}$$

The intersection of two even cut sets need not be even. If

$$\omega_3(Q) = |\delta(Q) \cap K_1 \cap K_2| \pmod 2,$$

then $\oplus_Q \omega_3(Q) = 0$, because every quotient edge is counted at both endpoints. Bad components therefore occur in pairs. The missing selection theorem must cancel these paired local defects.

5.1 Exact exchange law

Theorem 5.3 (odd-kernel exchange). *Let* $T' = T - a + b$*, where* $b \notin T$ *and* $a \in C_T(b)$*. Then*

$$K(T') = \begin{cases} K(T), & a \notin K(T), \\ K(T) \triangle C_T(b), & a \in K(T). \end{cases} \tag{5.6}$$

Proof. In the old cotree coordinates, the completion $F(T')$ is either $F(T)$ or $F(T) \triangle C_T(b)$. The new cotree contains a, so the first choice occurs exactly when $a \in F(T)$. Apply $F(T) = E - K(T)$. $\square$

If K_3 is fixed and exchanges in the first two trees give

$$K_1' = K_1 \triangle \alpha C_1, \qquad K_2' = K_2 \triangle \beta C_2$$

with $\alpha, \beta \in \{0, 1\}$, then edge indicators satisfy

$$\mathbf{1}_{K_1' \cap K_2'} = \mathbf{1}_{K_1 \cap K_2} + \alpha \mathbf{1}_{C_1 \cap K_2} + \beta \mathbf{1}_{C_2 \cap K_1} + \alpha\beta \mathbf{1}_{C_1 \cap C_2}. \tag{5.7}$$

The quotient-boundary target is therefore bilinear. Ordinary linear cycle-space invariance does not imply (5.4).

5.2 One parity element

The odd kernel is nonlinear as a function of a tree, but it is a standard fundamental circuit after one binary extension. This observation makes the remaining selection problem accessible to matroid language without changing its semantics.

Let G now be any connected loopless graph of even order n. Delete the row of a vertex ρ from its binary vertex-edge incidence matrix, obtaining

$$A \in \mathbb{F}_2^{(n-1) \times E(G)}.$$

Put $p = \mathbf{1} \in \mathbb{F}_2^{n-1}$, and let $\widetilde{M}$ be the binary matroid represented by $[A \mid p]$.

Theorem 5.4 (parity-element representation)**.** *For every spanning tree T,*

$$C_{\widetilde{M}}(T,p) = K(T) \cup \{p\}. \tag{5.8}$$

Equivalently, for every $e \in T$,

$$\mathbf{1}_{e\in K(T)} = \det A_{T-e+p} \pmod 2, \tag{5.9}$$

where A_{T-e+p} replaces the column of e in the tree basis by p.

Proof. The square matrix A_T is invertible. If $z = A_T^{-1}p$, then the nonroot incidence equations say that the edge set $\operatorname{supp}(z) \subseteq T$ has odd degree at every nonroot vertex. The number of odd-degree vertices is even, while n is even, so the root degree is odd as well. The uniqueness part of Theorem 5.1 gives $\operatorname{supp}(z) = K(T)$. Thus

$$p = \sum_{e\in K(T)} A_e$$

is the unique dependence of p on the basis T, and its support is the fundamental circuit (5.8). Cramer's rule over $\mathbb{F}_2$ gives (5.9); the determinant of a binary tree-incidence basis is one. $\square$

Parallel graph edges are distinct equal columns, so the statement treats them as distinct parallel matroid elements. A graph loop is a zero column and belongs to neither a spanning tree nor its odd kernel.

5.3 Root deletion and three cographic bases

The vertex-star case has structure absent from a generic type-A defect set. Let r be the chosen cubic vertex, put $D = \delta(r)$, and choose the deleted incidence row to be the row of r. Summing all rows of $[A \mid p]$ gives the row-space cocycle

$$D \cup \{p\}. \tag{5.10}$$

Indeed, an edge outside D has two nonroot incidences, an edge in D has one, and the p-column has $n-1$, hence an odd number, of ones. Circuit-cocycle orthogonality applied to (5.8) says that $|K(T) \cap D|$ is odd.

Proposition 5.5 (cographic-base form of a star fibre)**.** *Let $H = G - r$. A star-fibre packing is equivalent to:*

1. *a partition $E(H) = R_1 \dot\cup R_2 \dot\cup R_3$ into three bases of $M^*(H)$; and*
2. *a bijection between the three sets R_i and the three neighbours u_i of r.*

Writing $S_i = E(H) - R_i$, the corresponding tree is $T_i = S_i + ru_i$. Moreover,

$$J_i = K(T_i) - ru_i$$

is the unique subset of the tree S_i with boundary

$$\partial_H J_i = V(H) - \{u_i\}. \tag{5.11}$$

Proof. Each tree in a star-fibre packing must use an edge of D. There are three trees and only three available root-edge copies, so each tree uses exactly one, say ru_i. Removing the leaf r leaves a spanning tree S_i of H. Every internal edge occurs in exactly two of the S_i, hence their complements R_i partition $E(H)$. Complements of spanning trees are precisely the bases of the cographic matroid.

Conversely, complements of three cographic bases are three spanning trees of H. Adding the three root spokes bijectively reconstructs the required multiplicities. The root spoke belongs to $K(T_i)$, since r is a leaf of T_i and has odd kernel degree. Removing it makes the degree at u_i even and leaves every other vertex of H odd, which proves (5.11). Uniqueness follows because a tree has no nonempty Eulerian edge-subset. □

This gives a clean sufficient strengthening of Corollary 5.2. For distinct i, j, k, require

$$K(T_j) \cap K(T_k) \subseteq \mathrm{cl}_{M(G)}(K(T_i)). \tag{5.12}$$

Because $K(T_i)$ is a forest, an edge is in its graphic closure exactly when its endpoints are in one $K(T_i)$-component. Thus every edge of the intersection becomes a loop after contraction, and (5.12) implies the exact even-boundary condition. In the cographic-base form,

$$J_j \cap J_k \subseteq \mathrm{cl}_{M(H)}(J_i). \tag{5.13}$$

The root is a leaf, so it cannot create a new path between two vertices of H; hence (5.12) and (5.13) are equivalent.

There is also a literal linear-algebra test. If a_e is the incidence column of $e \notin T_i$, then

$$e \in \mathrm{cl}(K(T_i)) \iff \mathrm{supp}(A_{T_i}^{-1} a_e) \subseteq \mathrm{supp}(A_{T_i}^{-1} p). \tag{5.14}$$

The left support is the T_i-path joining the endpoints of e, and Theorem 5.4 identifies the right support with $K(T_i)$. Condition (5.12) is stronger than the desired parity condition: Section 6.4 gives both a sharp generic type-A countermodel and a dual-certified vertex-star countermodel.

6 Human-sized obstructions to stronger lemmas

The next results do not obstruct the exact component-parity target. They identify proof strategies that lose essential flexibility.

6.1 Petersen forbids disjoint odd kernels

Proposition 6.1 (Petersen no-go)**.** *For every triple of spanning trees of the Petersen graph and every $i \neq j$,*

$$K(T_i) \cap K(T_j) \neq \varnothing. \tag{6.1}$$

In particular, the quotient condition (5.4) cannot be proved by always making a pure-coordinate class empty.

Proof. Let K, L be edge-disjoint spanning subgraphs of a cubic graph, both having odd degree at every vertex. At each vertex, $\deg_K(v) + \deg_L(v) \leq 3$, while both summands are positive and odd. Thus both degrees are one: K and L are disjoint perfect matchings. Their union is a 2-factor whose cycles alternate between the matchings, so all its cycles are even.

The Petersen graph has no even 2-factor. In its outer-pentagon, inner-star presentation, a perfect matching uses an odd number of spokes. Three spokes are impossible: the two unmatched outer indices would have to differ by one modulo five, while the same inner indices would have to differ by two. With one or five spokes, the complementary 2-factor is two 5-cycles. Theorem 5.1 now proves (6.1). □

Exhaustion gives a useful positive control. For each of the ten Petersen stars there are 4,416 ordered packings, no packing has disjoint odd-kernel pairs, but 384 packings satisfy (5.4) in some coordinate.

6.2 The cube refutes a fixed-third-tree repair

Use the following edge order for the cube:

$$\begin{array}{llll} 0:04 & 1:05 & 2:06 & 3:14 \\ 4:15 & 5:17 & 6:24 & 7:26 \\ 8:27 & 9:35 & 10:36 & 11:37. \end{array}$$

Take root 0, so the star is $\{0,1,2\}$, and fix

$$T_3 = \{0,3,4,7,8,9,11\}, \qquad K_3 = \{0,4,7,11\}. \tag{6.2}$$

The K_3-components are

$$\{0,4\}, \quad \{1,5\}, \quad \{2,6\}, \quad \{3,7\}. \tag{6.3}$$

Proposition 6.2 (fixed-base no-go)**.** *The tree T_3 in (6.2) occurs in* 16 *ordered partitions of the cube star fibre, but none of its completions T_1, T_2 satisfies (5.4). The same star fibre has a parity-good jointly chosen triple.*

Proof. Normalize by putting root edge 1 in T_1 and root edge 2 in T_2. Swapping the first two trees gives the other eight completions. For all eight normalized rows,

$$K_1 = \{1,5,6,10\}.$$

The complete residual enumeration is:

T_1	T_2	K_2	$K_1 \cap K_2$
1 3 4 5 6 7 10	2 5 6 8 9 10 11	2 5 6 9	5 6
1 3 4 5 6 10 11	2 5 6 7 8 9 10	2 5 6 9	5 6
1 3 5 6 7 9 10	2 4 5 6 8 10 11	2 4 6 11	6
1 3 5 6 9 10 11	2 4 5 6 7 8 10	2 4 6 7 8 10	6 10
1 4 5 6 7 8 10	2 3 5 6 9 10 11	2 5 6 9	5 6
1 4 5 6 8 10 11	2 3 5 6 7 9 10	2 5 6 9	5 6
1 5 6 7 8 9 10	2 3 4 5 6 10 11	2 4 6 11	6
1 5 6 8 9 10 11	2 3 4 5 6 7 10	2 3 4 5 7 10	5 10

In the quotient by (6.3), edge 5 joins the second and fourth components, edge 6 joins the third and first, and edge 10 joins the fourth and third. Each of

$$\{5,6\}, \quad \{6\}, \quad \{6,10\}, \quad \{5,10\}$$

has an odd-degree quotient vertex. This proves failure for all sixteen ordered completions.

Joint choice succeeds, for example, with

$$T_1 = \{1,3,4,7,8,9,11\},$$
$$T_2 = \{2,5,6,8,9,10,11\},$$
$$T_3 = \{0,3,4,5,6,7,10\}.$$

Here $K_1 = \{1,3,7,11\}$, $K_2 = \{2,5,6,9\}$, and their intersection is empty. □

Of the 132 extendible third trees in this cube star fibre, 12 are bad in the sense of Proposition 6.2. A proof must co-optimize all three bases rather than freeze an arbitrary T_3.

6.3 Perfect forests cannot be extended blindly

A perfect forest is a spanning all-odd forest whose components are induced. Scott proved that every connected even-order graph has one [15]; Gutin gave a short linear-algebra proof [4]. Choosing three such forests before choosing the trees is tempting, because any spanning tree containing an all-odd forest has it as its odd kernel. The simultaneous extension step is nevertheless false.

Proposition 6.3 (ten-copy rank obstruction)**.** *There is a 3-edge-connected cubic graph G, a root-star capacity vector, and three capacity-respecting perfect matchings K_1, K_2, K_3 satisfying $K_1 \cap K_2 = \varnothing$, which do not extend to a three-tree partition of the star fibre.*

Proof. Use the labelled ten-vertex Möbius ladder

$$\begin{array}{lllll} 0:05 & 1:06 & 2:07 & 3:15 & 4:17 \\ 5:18 & 6:25 & 7:28 & 8:29 & 9:36 \\ 10:37 & 11:39 & 12:46 & 13:48 & 14:49. \end{array}$$

At root 0, choose

$$\begin{aligned} K_1 &= \{0, 4, 7, 11, 12\}, \\ K_2 &= \{1, 5, 6, 10, 14\}, \\ K_3 &= \{2, 5, 6, 11, 12\}. \end{aligned} \tag{6.4}$$

Each is a perfect matching, the root edges occur once, every other edge occurs at most twice, and $K_1 \cap K_2 = \varnothing$.

After reserving these copies, let X be the ten residual copies

$$3a, 3b, 7, 8a, 8b, 9a, 9b, 13a, 13b, 14. \tag{6.5}$$

After contracting K_i, the graphic-matroid rank of X is three for each i. This is visible without matroid terminology: adding the underlying edges of X to the five matching components leaves exactly two vertex components in each coordinate:

$$\begin{array}{r|l} K_1 + X & \{0,1,5,7\} \mid \{2,3,4,6,8,9\} \\ K_2 + X & \{0,3,6,7\} \mid \{1,2,4,5,8,9\} \\ K_3 + X & \{0,7\} \mid \{1,2,3,4,5,6,8,9\}. \end{array}$$

Thus an acyclic extension in coordinate i uses at most three members of X. Three tree extensions could use at most nine, but the residual partition must assign all ten members of X. This contradiction proves nonextension. □

The obstruction concerns a preselected forest triple. Another perfect forest triple in the same graph and fibre does extend and meets the parity target. Hence Scott's theorem remains potentially useful only if its forests are selected jointly with all matroid-union rank inequalities.

6.4 Kernel closure is false

The closure strengthening (5.12) fails both for a generic type-A defect set and, at a slightly larger order, for a vertex star. The two failures use different certificate styles.

Proposition 6.4 (exact 12-vertex closure no-go)**.** *The simple 3-edge-connected cubic graph with graph6 record*

`K?‘@E‘gFCKEO`

has two automorphic type-A fibres, with defect triples

$$D_0 = \{0,4,11\}, \qquad D_1 = \{1,3,10\}, \tag{6.6}$$

such that neither fibre has a tree packing satisfying (5.12) in any coordinate. Each fibre nevertheless has 7,704 *ordered packings satisfying the exact quotient-parity condition in one coordinate.*

The edge order decoded from the graph6 record is

$$\begin{array}{rrrrrr} 0:04 & 1:15 & 2:26 & 3:07 & 4:17 & 5:37 \\ 6:18 & 7:28 & 8:48 & 9:39 & 10:49 & 11:59 \\ 12:0\,10 & 13:5\,10 & 14:6\,10 & 15:2\,11 & 16:3\,11 & 17:6\,11. \end{array} \tag{6.7}$$

The edges $0,10,11,1,4,3$ form the six-cycle $0,4,9,5,1,7,0$, and D_0, D_1 are its alternating matchings.

This proposition is an exhaustive finite statement, rather than a short structural theorem. Its completeness can nevertheless be audited directly. A standard-library Python checker finds exactly 8,640 spanning trees. If D is one of (6.6), $M = E - D$, and the first two tree masks are fixed, multiplicity forces

$$T_3 = ((T_1 \triangle T_2) \cap M) \cup (D - (T_1 \cup T_2)). \tag{6.8}$$

The checker enumerates all ordered pairs among the 8,640 trees, uses (6.8), and retains the pair exactly when the forced third mask is also a spanning tree. It obtains 355,392 ordered packings per fibre. Direct reconstruction of every odd kernel gives the complete histogram

closure-good coordinates	parity-good coordinates	packings
0	0	347,688
0	1	7,704.

(6.9)

For a positive control, one parity-good packing for D_0 is

$$\begin{aligned} T_1 &= \{0,1,2,3,4,5,6,7,10,14,15\}, \\ T_2 &= \{1,2,5,8,9,10,11,12,13,16,17\}, \\ T_3 &= \{3,6,7,8,9,12,13,14,15,16,17\}. \end{aligned} \tag{6.10}$$

Its odd kernels are

$$\begin{aligned} K_1 &= \{1,3,4,5,6,10,14,15\}, \\ K_2 &= \{1,2,5,8,9,12,16\}, \\ K_3 &= \{3,6,7,8,9,13,17\}. \end{aligned}$$

These displayed sets permit a direct human check of the positive claim. The universal negative count in (6.9) depends on the enumerator and its independent semantic implementation; it has no LRAT certificate. The checker SHA-256 is

`a0d38e436994449f2c6d568b1cb6b3975d47754f94e001d94e8215d6e6817c7f.`

Thus Proposition 6.4 already blocks a closure theorem for arbitrary type-A fibres. It does not yet use the special star cocycle (5.10).

Theorem 6.5 (dual-certified vertex-star closure no-go)**.** *There is a simple 3-edge-connected cubic graph G of order* 16 *and a vertex r such that no packing in the vertex-star fibre at r satisfies (5.12) in any coordinate. The same fibre has a compatible pair labelling supported on five points and hence satisfies the exact quotient-parity target.*

The graph6 record and root are

$$\texttt{O??CA?\_ceOGgH\_F?AK@P?}, \qquad r = 13, \tag{6.11}$$

and the root-star edge indices are $10, 13, 17$ in column-major graph6 edge order. An independent parser verifies that the graph is simple, cubic, 3-edge-connected, and even 3-vertex-connected.

For auditability, we describe the complete certificate semantics. The CNF has 744 variables and 101,037 clauses. Variables $t_{e,i}$ and $k_{e,i}$ encode the trees and odd kernels. Edge rows impose multiplicity one on the three root edges and multiplicity two elsewhere. For each coordinate, one cut clause for every nonempty subset of $V - \{0\}$ forces connectivity. The total multiplicity is $3(16-1) = 45$; three connected spanning subgraphs use at least 15 edges each, so every coordinate has exactly 15 edges and is a spanning tree.

Clauses $k_{e,i} \Rightarrow t_{e,i}$, together with odd incidence parity at every vertex, make $k_{\cdot,i}$ the unique all-odd subset of that tree, namely $K(T_i)$. A variable q_e is constrained to be equivalent to $k_{e,0} \wedge k_{e,1}$. Conditional path variables then require, when $q_e = 1$, a boundary-∂e edge set supported inside $K(T_2)$. Since $K(T_2)$ is a forest, such a set exists exactly when the endpoints of e lie in one $K(T_2)$-component. The formula therefore asks for (5.12) in coordinate two. Tree-coordinate symmetry makes it satisfiable exactly when some coordinate works.

CaDiCaL produced an LRAT proof of UNSAT. Both the C `lrat-check` checker and the formally verified CakeML `cake_lpr` checker accept it:

$$\begin{array}{ll} \text{CNF SHA-256} & \texttt{ccd8ef9504ef725dc02be9234992cbc9e238f1c12ed39ad3068645c73e3c5baf}, \\ \text{LRAT SHA-256} & \texttt{ffcd95bc48db7c5be222e05b86aa72cf0828ef01e45e7cf45e9a92cab76aa3a5}. \end{array} \tag{6.12}$$

The independent wrapper checks those bytes and both proof-checker outcomes. It also replays a witness produced by a different support-five encoding: three literal spanning trees in this fibre yield exact component parity in coordinate two and pair labels supported on $\{0, 4, 5, 6, 7\}$. Thus the theorem refutes only the closure strengthening, not the star-fibre parity target or FiveCDC.

An independent solver-free C++ enumeration uses Proposition 5.5. It finds 16,200 cographic bases of $G - r$, 158,976 unordered three-base partitions, and 5,723,136 ordered star packings after all coordinate and spoke assignments. Its complete histogram is

closure-good coordinates	parity-good coordinates	packings
0	0	5,682,672
0	1	40,464.

(6.13)

This duplicates the UNSAT conclusion by direct enumeration and shows that exact parity is not rare in the same fibre.

A complete producer scan of all 4,060 connected simple cubic graphs of order 16, all 64,960 roots, and all 45,248 feasible star fibres finds exactly this one closure failure. Together with the complete positive star census through order 14, this makes order 16 minimal within the connected simple cubic star-fibre domain. The witness contains the triangle $0, 6, 9$; hence it does not refute the closure strengthening on strict snarks or even on the girth-at-least-five subclass. The frozen stream has SHA-256

$$\texttt{889d0e5857b62e082f3fcad0a929e7c0d92032a8e675f4d2ff9f40272ea43d46}.$$

This order-16 census is a producer computation; the UNSAT claim for the one displayed fibre has the stronger dual-checked certificate above.

Triangle contraction and an infinite auxiliary family

The graph6 record has a useful structural explanation, and the resulting statement has a short human proof.

Theorem 6.6 (triangle-contraction theorem)**.** *Let $G^{\triangle}$ be a simple cubic graph, let Δ be a triangle, and let G be obtained by contracting Δ to one vertex. Fix a root $r \notin V(\Delta)$. Every closure-good star packing of $(G^{\triangle}, r)$ contracts to a closure-good star packing of (G, r). Consequently, triangle expansion at any nonroot vertex preserves failure of (5.12).*

Proof. Every triangle edge has multiplicity two, so the three trees have six triangle-edge occurrences in total. A tree contains at most two edges of a triangle. Hence each tree T_i contains exactly two. Contracting those two edges produces a spanning tree $\bar{T}_i$ of G, and all external multiplicities are unchanged.

For an edge $e \notin E(\Delta)$, all three triangle vertices lie on one side of $T_i - e$. Contraction removes two vertices from that side, so its order has the same parity. Therefore

$$e \in K(T_i) \quad \Longleftrightarrow \quad e \in K(\bar{T}_i) \qquad (e \notin E(\Delta)). \tag{6.14}$$

Suppose the packing upstairs is closure-good in coordinate k, and let $e \in K(\bar{T}_i) \cap K(\bar{T}_j)$. By (6.14), its corresponding external edge lies in $K(T_i) \cap K(T_j)$, so its endpoints are joined by a $K(T_k)$-path. Contracting the triangle maps that path to a $K(\bar{T}_k)$-walk and hence to a path. Thus the contracted packing is closure-good. The final assertion is the contrapositive. □

Triangle expansion preserves simplicity and cubicity. It also preserves 3-edge-connectivity: any cut of size at most two that does not split the new triangle contracts unchanged; if it splits one vertex from two, or two from one, replacing the two crossing triangle edges by the two corresponding external edges gives a cut of no larger size after contraction. Starting from Theorem 6.5 and repeatedly expanding any nonroot vertex therefore gives an infinite family of simple cubic 3-edge-connected rooted closure countermodels, of orders

$$16 + 2t, \qquad t \geq 0. \tag{6.15}$$

This is an infinite family only for the discarded sufficient strengthening (5.12). The outside-root hypothesis is essential to this proof. If the root lies inside the expanded triangle, its three triangle edges have total multiplicity four rather than six, so at least one coordinate cannot contract to a tree.

The local lifting law explains the obstruction. Expand a contracted tree vertex z to a triangle abc, and suppose the lifted tree uses the path $a - b - c$. Let b_x indicate whether the external edge at x belongs to the downstairs odd kernel. Odd degree at z gives $b_a + b_b + b_c = 1$, and odd degree at the three lifted vertices forces

$$\mathbf{1}_{ab \in K(T)} = 1 + b_a, \qquad \mathbf{1}_{bc \in K(T)} = 1 + b_c \pmod 2. \tag{6.16}$$

If all three external traces equal one, neither new path edge belongs to the lifted kernel, so one downstairs component splits into three. In a literal 14-vertex quotient packing, the standard checker reconstructs all six bijections between tree coordinates and omitted triangle edges: the downstairs closure vector is $(0, 0, 1)$, while every lifted vector is $(0, 0, 0)$.

Finally, the 16-vertex countermodel has three disjoint triangles

$$\{0, 6, 9\}, \qquad \{1, 7, 11\}, \qquad \{2, 8, 15\}.$$

Contracting all three gives the Petersen graph. Relative to the image of the root, its six distance-two vertices induce a 6-cycle, and the expanded vertices are one alternating class. The two alternating classes at each of ten roots form one automorphism orbit, giving $10 \cdot 2 = 20$ rooted Petersen triple-expansions, all rooted-isomorphic to the certified pair. The independent structural checker verifies the contraction, the six-lift calculation, and this Petersen description. The contraction theorem itself is the human argument above.

Exact component parity is triangle-invariant

The preceding infinite family concerns only the false closure strengthening. The exact target behaves differently. We first record a linear form that makes the distinction transparent. Write ∂X for the set of vertices of odd degree in an edge set X.

Lemma 6.7 (Eulerian completion in a forest)**.** *Let K be a forest and let $A \cap K = \varnothing$. The set A has even boundary on every component of K if and only if there is a unique $P \subseteq K$ such that $A \triangle P$ is Eulerian.*

Proof. The condition is $\partial P = \partial A$. On each tree component of K, the binary incidence map is injective and its image is exactly the even-cardinality vertex sets. This proves both existence and uniqueness. □

Consequently, for a star packing T_0, T_1, T_2, with $K_i = K(T_i)$, the exact coordinate-two target is

$$\exists P \subseteq K_2 \quad (K_0 \cap K_1) \triangle P \text{ is Eulerian.} \tag{6.17}$$

The triple intersection is empty because $K_i \subseteq T_i$ and no edge belongs to all three trees. Thus the disjointness hypothesis in Lemma 6.7 applies. Unlike (5.12), condition (6.17) allows an entire cycle in the quotient by K_2; it does not require each common edge to become a quotient loop.

Theorem 6.8 (exact nonroot triangle invariance)**.** *Let $G^{\triangle}$ be obtained from a cubic graph G by expanding a vertex z to a triangle, and let the star root r lie outside that triangle. The star fibre at r contains a packing satisfying exact component parity in $G^{\triangle}$ if and only if the corresponding contracted fibre contains one in G.*

Proof. As in Theorem 6.6, each tree contains two triangle edges, contracts to a spanning tree, and preserves membership of every external edge in its odd kernel.

Suppose first that (6.17) holds upstairs. Contract the Eulerian set $(K_0 \cap K_1) \triangle P$ and discard triangle loops. Contraction preserves even degrees, and external odd-kernel membership is unchanged. The result is

$$(\bar{K}_0 \cap \bar{K}_1) \triangle \bar{P}, \qquad \bar{P} \subseteq \bar{K}_2,$$

so exact parity holds downstairs.

For the converse, fix a downstairs witness $\bar{P} \subseteq \bar{K}_2$. Name the expanded vertices a, b, c. Let $b_i \subseteq \{a, b, c\}$ record which external edges at the contracted vertex belong to $\bar{K}_i$, and let $p \subseteq b_2$ record the external trace of $\bar{P}$. The local data obey

$$|b_i| \equiv 1 \pmod 2, \qquad b_0 \cap b_1 \cap b_2 = \varnothing, \qquad |(b_0 \cap b_1) \triangle p| \equiv 0 \pmod 2. \tag{6.18}$$

The first condition is odd degree in $\bar{K}_i$, the second follows from the tree multiplicities, and the third is Eulerian parity of the downstairs witness at z.

A legal lift chooses omitted triangle edges (o_0, o_1, o_2) as a permutation of ab, bc, ca. Let x_i be the triangle-edge part of the lifted K_i. It is the unique subset of the two included triangle edges with

$$\partial x_i = \{a, b, c\} \triangle b_i. \tag{6.19}$$

Put $z_{\text{ext}} = (b_0 \cap b_1) \triangle p$ and $q = x_0 \cap x_1$. It is enough to find $y \subseteq x_2$ such that

$$\partial y = z_{\text{ext}} \triangle \partial q, \tag{6.20}$$

because then $P = \bar{P} \cup y$ witnesses (6.17) upstairs.

There are only seven cases up to permuting a, b, c and swapping coordinates zero and one. In the following table, + denotes symmetric difference. Each displayed x_i is obtained from (6.19), and direct substitution verifies (6.20).

(b_0, b_1, b_2)	p	(o_0, o_1, o_2)	(x_0, x_1, x_2)	y
(a, a, b)	b	(ab, bc, ca)	$(bc, ab{+}ca, ab{+}bc)$	ab
(a, b, a)	0	(ab, bc, ca)	(bc, ca, bc)	0
(a, b, c)	0	(ab, bc, ca)	(bc, ca, ab)	0
(a, b, abc)	0	(ab, bc, ca)	$(bc, ca, 0)$	0
(a, b, abc)	$a{+}b$	(bc, ca, ab)	$(ab{+}ca, ab{+}bc, 0)$	0
(a, b, abc)	$a{+}c$	(bc, ab, ca)	$(ab{+}ca, ca, 0)$	0
(a, abc, b)	b	(ab, bc, ca)	$(bc, 0, ab{+}bc)$	ab

(6.21)

For completeness, conditions (6.18) admit 60 labelled quadruples. The seven displayed orbits have sizes

$$6, 12, 6, 6, 6, 12, 12,$$

whose sum is 60. Hence the table covers every local trace and proves that some legal lift preserves exact parity. □

This proof is finite but human-checkable: the three equations (6.18)–(6.20), the seven rows, and the orbit-size sum are the entire local argument. An independent standard-library checker enumerates all 60 traces and confirms seven orbits with no failed lift. The numbers of traces having one, two, three, or four good lifts are respectively $18, 12, 6, 24$. The human note and checker have SHA-256, respectively,

`96259e65072b5a8c03f572e028a650f6e075f787c66e7e59917630c2b22c1008`,

`e3026b8acdf95555fad33a5a2871174b497a154151f5e0f4fde642d31053bbe7`.

For the simple-graph reduction, a small caveat must be closed. In a simple 3-edge-connected cubic graph other than K_4, the three vertices of a triangle have distinct external neighbours. If exactly two shared an external neighbour, the triangle together with that neighbour would expose a cut of size two; if all three shared it, the connected cubic graph would be K_4. Triangle contraction therefore remains simple. It remains 3-edge-connected because any cut of size at most two downstairs lifts by placing the whole triangle on the contracted vertex's shore. The K_4 star fibres are checked directly.

It follows that every triangle in a prescribed-root minimal simple 3-edge-connected obstruction to the exact star-fibre statement must contain the root. For the unrooted existence problem, one may choose a root outside a triangle and contract it. This does not handle a preassigned root inside the triangle and does not prove the triangle-free case. Most importantly, Theorem 6.8 shows that the certified 16-vertex closure countermodel and its infinite expansion family are not obstructions to the exact target: closure descends but can fail to lift, whereas exact parity both descends and lifts.

6.5 A four-cycle warning: statewise local lifting fails

The triangle theorem does not extend by simply replacing the triangle by a square. The distinction is already visible without SAT.

Proposition 6.9 (square-local statewise lifting no-go). *There are a simple planar 3-edge-connected cubic graph H, a nonroot independent edge pair AC, BD, and an exact-good star-fibre state $\mathcal{T}$ such that, after replacing AC, BD by the square*

$$Aa, Bb, Cc, Dd, ab, bc, cd, da,$$

no star-fibre lift that fixes $\mathcal{T} - \{AC, BD\}$ outside the gadget is exact-good in any of the three coordinate directions.

The labelled downstairs graph is `G?zTb_`, with root 0, and the displayed state in the reproduction note has profile

$$(2, 2, 0).$$

Every one of the eight gadget edges has multiplicity two, so a local lift is specified by its omitted tree on each edge. Exhausting all $3^8 = 6561$ omitted-owner words leaves exactly 72 legal three-tree lifts. Their complete profile histogram is

profile	$(2,2,2)$	$(2,2,4)$	$(2,4,2)$	$(4,2,2)$	$(4,4,2)$
count	38	12	2	2	18.

Thus no local lift is good even after changing the output coordinate.

This does *not* refute the whole-fibre existential square reduction. The expanded graph has a separately displayed star-fibre state of profile $(0, 4, 4)$. More sharply, the same downstairs graph, root, and smoothing pair have another good state, of profile $(0, 2, 2)$, with a good square-local lift. The proposition therefore rejects only the quantifier "lift this already chosen good state locally." A valid square induction would have to coordinate the state selection with the lift, and the exact whole-fibre implication remains open.

The exact finite boundary has also been checked below this first failure. For both simple 3-edge-connected cubic graphs of order six, every good downstairs state has an any-coordinate good local lift for every eligible smoothing pair. At order eight, a complete `geng` census of the four simple 3-edge-connected cubic graphs, all eight roots, and all twenty-one eligible pairs checks 672 graph/root/pair instances. In every instance some good downstairs state has a good local lift; only 688 state attempts are needed in total. Thus order eight is the first fixed-state failure but supplies no counterexample to the existential square implication.

There is an exact criterion at the level of one already chosen five-cover. Label each edge by the two cover coordinates containing it. Suppose the deleted edges AC and BD have labels P and Q, respectively, and hold all labels outside the square fixed. Terminal parity forces the four attachment labels to be P, Q, P, Q. If X is the label on ab, then parity around the square forces

$$X, \qquad X \triangle Q, \qquad X \triangle P \triangle Q, \qquad X \triangle P. \tag{6.21}$$

All four sets in (6.21) have size two if and only if

$$P = Q \quad \text{or} \quad P \cap Q = \varnothing. \tag{6.22}$$

For $P = Q$, choose X meeting P once; for disjoint P, Q, choose one point of each. If $P = \{p, q\}$ and $Q = \{p, s\}$, the three weight conditions would give

$$x_p + x_q = x_p + x_s = x_q + x_s = 1,$$

which is impossible. This proves (6.22). The criterion is deliberately for a fixed outside five-labelling; it does not identify which exact-good tree state should be selected.

A standard-library, solver-free census tests that remaining selection quantifier on every simple 3-edge-connected cubic graph of orders ten and twelve. For every labelled root and every eligible independent edge pair, it enumerates the entire star fibre and accepts only after reconstructing a good downstairs state and a good square-local lift. The exact totals are

order	3-edge-connected graphs	roots	edge-pair instances	failures
10	14	140	6,300	0
12	57	684	53,352	0.

The search required 6,923 and 66,696 good-state/pair attempts, respectively; in particular, the first good state need not always be the one that lifts. The witness-corpus SHA-256 values are

10 : `1ae75f2864163e8ec6b04c0d92f012f00375ab55f92cc1284ad8233d4c42ab8f`,

12 : `b6b5420643799ddfe9ab3f252b0447c7de7704b27ddd8e75eaed39613b567b4e`.

At order fourteen, a C++ search writes an explicit pair of downstairs and lifted tree triples for every instance. A separately written standard-library checker regenerates the canonical `geng` stream, independently filters three-edge-connectivity, requires exactly one row for each graph, root, and independent edge-pair key, and reconstructs all six trees, their multiplicities, outside agreement, and both exact component-parity tests. It accepts

order	3-edge-connected graphs	roots	edge-pair witnesses	failures
14	341	4,774	572,880	0.

The corpus uses 24,268 distinct downstairs states and has SHA-256

`26dfe990a6655170f5fdcb6faf1424db279b57e8d094269417b7330fb5affd41`.

Thus any countermodel to this stronger local-selection statement in the simple 3-edge-connected cubic class has downstairs order at least sixteen. This is a finite certificate census, not a proof of the whole-fibre implication.

7 A certified obstruction to direction thinning

A much broader sufficient rule would try to choose a star-fibre state in which some nonzero flow value occurs at most once. Such thinning forces at least three missing pairs in the corresponding perfect matching of K_8 and hence gives a five-colourable co-occurrence graph. The rule is false.

Theorem 7.1 (certified 34-vertex countermodel)**.** *There is a simple, cubic, 3-edge-connected 34-vertex graph and a vertex-star fibre such that for every state (T_1, T_2, T_3) in the fibre and every $d \in \mathbb{F}_2^3 - \{0\}$,*

$$|\phi^{-1}(d)| \geq 2. \tag{7.1}$$

The graph6 encoding is

```
as???SK????A?A?C_@_?B?@_??G??_I?GA_AA????_??@?????B??@_?o??????_C??CGO??B@????a????__??C@O??A?_
```

In column-major edge order, edges $0, 1, 2$ are the star at vertex 0. Their tree multiplicities are one and every other edge has multiplicity two.

7.1 Certificate semantics

The CNF has 4,444 variables and 251,803 clauses. For each tree coordinate, every nonroot vertex chooses one parent edge, the root chooses none, and unary levels strictly decrease from child to parent. Thus the selected 33 edges form a spanning tree, and every spanning tree has such an encoding. Edge rows impose the prescribed fibre.

Completion variables $f_{e,i}$ satisfy

$$e \notin T_i \Longrightarrow f_{e,i} = 1, \qquad \sum_{e \in \delta(v)} f_{e,i} = 0 \pmod 2. \tag{7.2}$$

Lemma 3.1 proves that these are the literal fundamental completions, not a relaxation. Tseitin variables define the seven possible values of $\phi(e)$. Selector clauses ask for one $d \neq 0$ that occurs on at most one edge. Consequently the CNF is satisfiable exactly when (7.1) is false.

CaDiCaL 3.0.1 [13] produced an LRAT certificate. The C checker `lrat-check` and the formally verified CakeML `cake_lpr` checker [16] both accept it.

CNF SHA-256	`8c96bd3d02685c12dca1eee78047459c65a60b0defdf87b68c000274dee9e6c0`,
LRAT SHA-256	`a9600d2cf2efa191f789438f811f116932d79ca0493f1522bc7e3b0ccee01ae4`.

The graph parser and certificate wrapper independently check simplicity, cubicity, connectivity after every one- and two-edge deletion, and the encoding invariants. This proves Theorem 7.1 subject to the usual trust boundary of the displayed semantic argument, the checked CNF bytes, and the proof-checker implementations.

7.2 The same fibre has exact five-point support

The obstruction is only to thinning. An independently replayed SAT witness in the same fibre has flow-value counts

$$(7, 5, 10, 7, 7, 7, 8)$$

and pair labels using exactly the five points

$$\{3, 4, 5, 6, 7\}.$$

All ten pairs among those points occur. Proposition 3.3 therefore gives a literal five-cycle double cover with no merge. A standard-library checker reparses the graph, verifies the trees and star multiplicities, recomputes all fundamental completions, and checks every pair xor and local parity equation. The witness SHA-256 is

`5516fa0bb0bf58f3bc3984de60b47e43f3f43c0cf45a28920e7f916d53d44063`.

Four-point support is impossible in this fibre: four affine points realize at most six nonzero pair differences, while Theorem 7.1 forces all seven differences to occur. Thus the minimum support here is exactly five.

8 Bounded exact computations

All statements in this section are finite computations. They are not inductive arguments and do not establish Problem 3.6.

8.1 Every fixed fibre through order fourteen

Nauty's canonical cubic-graph generation methods [2] were used to enumerate every connected simple bridgeless cubic graph of orders $4, 6, 8, 10, 12, 14$. For each type-A/type-B placement, a certificate-checking SAT encoding searched for three trees in the fixed fibre and a compatible labelling on at most five points. Every returned model was checked inside the producer by direct tree traversal, recomputation of the completions, pair xor, and point parity.

$\lvert V \rvert$	graphs	placements	feasible	five-point SAT
4	1	50	50	50
6	2	312	312	312
8	5	1,760	1,608	1,608
10	18	11,970	10,725	10,725
12	81	90,882	77,508	77,508
14	480	840,000	714,001	714,001
total	587	944,974	804,204	804,204

The frozen report has SHA-256

`482e2ee028aa1663f2b91b71569422ab59c1033d51601ca52fac581b173872f1`.

An independent structural verifier checks this hash, the canonical graph set, graph premises, pattern counts, source hashes, row totals, and the equality of feasible and successful counts. It does not retain and replay all 804,204 literal models. Accordingly, this census is exactly rerunnable but has a larger trust surface than the order-44 witness archive below.

8.2 Order thirty-eight strong snarks

The stronger canonical support

$$S_0 = \{0, 4, 5, 6, 7\} \tag{8.1}$$

was imposed in every vertex-star fibre of two complete retained order-38 strong-snark lists:

class	graphs	star fibres	SAT in S_0
girth at least 5, cyclic connectivity 4	298	11,324	11,324
girth 4, cyclic connectivity 4	407	15,466	15,466
total	705	26,790	26,790

The frozen result JSON has SHA-256

`9322f3dfa82d42966873cbb49f93dd328bfd1d57d61853a84db0ac21e2604d96`.

The producer performed direct semantic checks on every returned model, but this run did not archive a separate literal witness file. It is therefore reported as bounded computational evidence, not as a certificate package of the same strength as Theorem 7.1.

8.3 A retained order-forty-four family

For all 31 graphs in the retained order-44, oddness-four, cyclic-connectivity-four source file, all 44 roots admit a labelling in the same fixed support S_0. This gives 1,364 star witnesses. Unlike the preceding censuses, every literal witness is retained.

An independently written standard-library verifier decodes graph6, checks the 44-vertex simple cubic graph and bridgelessness, checks all three trees and the star multiplicities, reconstructs the fundamental completions, checks the emitted flow values, and checks every supported pair and every local parity equation. Its result is

Witness JSONL SHA-256
`3ff25437d5e32fff7c60ceca695a6476d6c7d2404c1e8551bf14987f228cd4b4`

Verification JSON SHA-256
`803055d00e601547ee7e59135089849796dc12367302a233822dde7075a1b560`

Normalized semantics SHA-256
`abb8bab730e873a6648d6e56987b6595f95162b2a3e7bd07d592b8d300790800`

This is not a complete order-44 snark census; it concerns exactly the stated 31-graph input family.

8.4 Symmetric Fano-minimum descent through order fourteen

For a star-fibre flow ϕ and each nonzero functional $h \in (\mathbb{F}_2^3)^*$, put

$$E_h = \{e : h(\phi(e)) = 0\}.$$

Theorem 4.2 assigns a leaf-parity bit $\omega_h(Q)$ to every component Q of E_h. Define

$$d_h(\phi) = |\{Q : \omega_h(Q) = 1\}|, \qquad d_{\min}(\phi) = \min_{h \neq 0} d_h(\phi). \tag{8.2}$$

Thus $d_{\min} = 0$ exactly when at least one of the seven associated five-point supports passes the component criterion.

Minimizing one preselected coordinate is not sufficient. On the 14-vertex graph

`M?AA@BORDGEOEOAo?`,

an explicit state has fixed-coordinate defect two and exactly 18 legal reciprocal exchange neighbours, all of fixed-coordinate defect four; the other 90 candidate swaps violate a tree condition. Yet its full seven-plane profile is

$$(2, 4, 2, 2, 0, 6, 2),$$

so that state is already good for another plane. One legal exchange has profile $(0, 2, 4, 4, 2, 2, 2)$. This exact countermodel therefore motivates the symmetric minimum rather than obstructing it.

Theorem 8.1 (finite symmetric descent)**.** *Let G be a simple 3-edge-connected cubic graph with $|V(G)| \leq 14$, fix a vertex r, and fix any assignment of its three spokes to the three trees. In the reciprocal-exchange graph of the corresponding star fibre, every connected component of every positive level set $d_{\min}^{-1}(k)$ has an exchange edge to a lower level.*

This is an exhaustive computer theorem. Here is the proof object and its trust boundary. Delete r, and let A_0, A_1, A_2 be the internal edges omitted by the three trees. The enumerator visits every ordered partition

$$E(G - r) = A_0 \,\dot{\cup}\, A_1 \,\dot{\cup}\, A_2$$

for which every spoke plus $E(G-r)-A_i$ is a spanning tree. It reconstructs each odd kernel by rooted odd-subtree sizes and then computes all seven instances of the leaf-bit system (4.7), not a proxy score. Every legal reciprocal swap is generated. Equal-level neighbours are joined in a disjoint-set structure, and each state with a lower-level neighbour marks its same-level component. The asserted failure condition is precisely a positive unmarked component.

Canonical graph generation, followed by explicit connectivity tests after every deletion of at most two edges, and one representative from every vertex-automorphism orbit gives:

$\|V\|$	3EC graphs	root orbits	states	failures
4	1	1	6	0
6	2	2	84	0
8	4	8	2,304	0
10	14	44	96,720	0
12	57	329	5,994,624	0
14	341	3,183	523,056,384	0
total	419	3,567	529,150,122	0

The maximum observed $d_{\min}$ is four. Graph automorphisms transport the complete state and exchange graphs. Permuting the three spoke assignments permutes flow coordinates and hence the seven functionals, so one assignment per root is sufficient.

The frozen row-level census has SHA-256

`f1d720265cc58a732cf2b58920b8dacc12f36191cecc731949622f5c80f4f2e0`.

An independently written verifier regenerates every canonical graph, 3-edge-connectivity filter, root orbit, row order, total, and digest. It does not re-enumerate the 529,150,122 states, so the C++ whole-state enumerator remains in the trust boundary. The theorem is finite: there is no reduction from a universal failure to order at most 14.

One targeted order-16 extension is especially relevant. In the vertex-star fibre of Theorem 6.5, the same whole-state enumerator checks all 953,856 states, again finds maximum $d_{\min} = 4$, and finds no trapped positive-level component. A separate Python implementation replays one explicit all-seven profile $(4,4,4,4,4,4,4)$ and all 147 incident exchange candidates; 25 are legal, of which 22 descend to symmetric defect two. The targeted result JSON has SHA-256

`90e1443fab33559480f7774bdb5dfffc904bfcd05efacf81ca845e55a4980626`.

This is an exact result for one rooted fibre, not a full order-16 census.

The exchange potential also admits a human update law. For a component Q of E_h, flow conservation implies that the four colour-class parities on $\delta(Q)$, for the four values in $h^{-1}(1)$, are equal. Their common value is the defect bit $\beta_h(Q)$; in particular every d_h is even. Under a reciprocal exchange between coordinates i, j, let Z_i, Z_j be the active fundamental cycles from (5.7), with an inactive cycle interpreted as empty. For every $f \neq 0$, put

$$Z_f = f(e_i)Z_i \triangle f(e_j)Z_j.$$

Then the complete zero-subgraph update is

$$E'_f = E_f \triangle Z_f. \tag{8.3}$$

Choose g, ℓ so that (g, ℓ, h) is a basis of the dual space. The new defect on every component Q' of $E_h \triangle Z_h$ is

$$\beta'_h(Q') = |\delta(Q') \cap (E_g \triangle Z_g) \cap (E_\ell \triangle Z_\ell)| \pmod 2. \tag{8.4}$$

When $Z_h = \varnothing$, the component partition is fixed. Expanding the intersection gives

$$\begin{aligned} R &= (Z_g \cap E_\ell) \triangle (E_g \cap Z_\ell) \triangle (Z_g \cap Z_\ell), \\ \beta'_h &= \beta_h + \partial_h R, \\ d'_h - d_h &= |\partial_h R| - 2|\operatorname{supp}(\partial_h R) \cap \operatorname{supp}(\beta_h)|. \end{aligned} \tag{8.5}$$

Equations (8.3)–(8.5) are identities, but do not force a favourable exchange.

Indeed, in the same order-16 fibre, the state with omitted-class masks

$$(857601, 1059160, 180390)$$

has profile $(4, 4, 4, 2, 4, 6, 4)$. Of 147 candidate reciprocal swaps, 23 are legal; 18 remain at $d_{\min} = 2$ and five rise to four. None descends. A kernel-inert same-level exchange followed by an active exchange reaches profile $(2, 2, 2, 2, 4, 6, 0)$.

The natural fixed-kernel exposure repair is also false. The state

$$(1722528, 307976, 66647)$$

has profile $(4, 4, 4, 4, 6, 2, 2)$. Again 23 of 147 swaps are legal and all remain at score two, but none preserves the ordered kernel triple. Its fixed-kernel realization component is therefore a singleton. An active same-level exchange changes the masks to $(1722504, 308000, 66647)$, after which a second exchange reaches $(1595528, 434976, 66647)$ with profile $(6, 4, 4, 4, 4, 4, 0)$. A standard-library checker reconstructs the graph, trees, kernels, all seven defects, both complete incident neighbourhoods, and both two-step paths. Thus only the full same-level-component statement in Theorem 8.1 survives these local simplifications.

Sorting all seven defects does not restore one-step monotonicity. On the 36-vertex graph and literal state recorded in the reproducibility note, the ordered profile is

$$(6, 6, 2, 4, 2, 2, 2), \qquad \sigma = (2, 2, 2, 2, 4, 6, 6).$$

Of $3 \cdot 17 \cdot 17 = 867$ reciprocal candidates, exactly 63 are legal: none has a lexicographically smaller sorted profile, 13 have the same profile, and 50 have a larger one. An independent standard-library checker exhausts the neighbourhood. Nevertheless, breadth-first search within $d_{\min} = 2$ finds a defect-zero state in two exchanges, after inserting 57 same-level states in deterministic order. Thus neither the scalar score nor its sorted refinement is a discrete-convex one-step potential; the full plateau-component statement still survives.

Nonroot triangle expansion also does not turn the finite theorem into an induction. The expanded fibre graph is the Cartesian product of the contracted fibre graph and the transposition Cayley graph of S_3, but the six lifted scores need not agree. One contracted good state has four good lifts and two lifts of score two. As a bounded control, all thirteen nonroot triangle expansions of the 14-vertex fixed-coordinate example were enumerated exactly: 17,297,280 states, maximum score four, and no trapped positive same-level component.

The graph product extends exactly to simultaneous expansions. If S is any set of nonroot vertices of the original loopless cubic graph, even with adjacent members, and G^S expands every member to a triangle, then

$$\mathcal{X}(G^S, r) \cong \mathcal{X}(G, r) \,\square\, \operatorname{Cay}(S_3, \mathcal{T})^{\square |S|}, \tag{8.6}$$

where $\mathcal{T}$ is the set of transpositions. Indeed, each expanded triangle has multiplicity two on all three edges, so the three trees omit distinct edges and give one S_3 coordinate. A reciprocal swap between external edges is precisely the contracted swap; a swap between two omitted edges of one triangle is a transposition. A mixed triangle/external swap, or a swap between two different triangles, makes one changed tree contain a complete triangle and is illegal. These are all possibilities and prove (8.6).

This additional product structure still does not control the defect. Exact simultaneous-lift searches on two adversarial order-16 states find no strict trap: all 3,780 two-triangle lifts of the first have at least fourteen equal-level neighbours when none descends, and all 3,780 lifts of the second have at least seven. Larger randomized plateau searches also escaped, but are explicitly exploratory. The product theorem is human; none of these finite searches proves the same-level-component statement.

For each tested state, Theorem 8.1 also gives a nonincreasing path to level zero. Follow equal-level exchanges to the marked boundary of the current component, take one descending exchange, and iterate. Finiteness of the nonnegative level proves termination.

8.5 The kernel-closure frontier

Before the countermodel in Theorem 6.5, the stronger closure condition (5.12) passes the following bounded domains:

domain	patterns	feasible	closure-good
vertex stars, connected simple cubic, through 14	8,392	5,646	5,646
all type A/B, connected simple cubic, through 12	110,127	90,203	90,201
retained order-44 stars	1,364	1,364	1,364

The two generic failures in the second row are exactly the automorphic fibres of Proposition 6.4. At order 16, however, the complete connected-simple-cubic stream has 4,060 graphs, 64,960 roots, 45,248 feasible star fibres, and exactly the one failure certified in Theorem 6.5. The retained 34-vertex strong snark passes at all 34 roots, and 50 deterministic random simple 3-edge-connected cubic graphs of order 50 pass at all 2,500 tested roots. Those latter rows are reproducible samples, not exhaustive families, and cannot rescue the universal closure statement.

The star-through-14 report and the type-A/B-through-12 report have SHA-256, respectively,

`e1042339b327659c7d6e9bbcfd46eb0576cd0f5644bf2cf3450b2e05998928a1`,

`80f68833c07bd555d3f7b5b8a2cc7b4a1b2bcc7d3b16c3472a4621101310e9d8`.

The producer replays returned models semantically. A separate standard-library verifier replays all 12,695 positive type-A/B witnesses through order ten, and Proposition 6.4 has its own independent complete enumerator. The order-16 failure has the independently checked CNF/LRAT package described in Theorem 6.5. The larger positive rows are bounded evidence with the producer inside their trust boundary.

9 The surviving theorem and failed shortcuts

The exact star-fibre target can now be stated without affine labels.

Open problem 9.1 (odd-kernel parity selection)**.** For every 3-edge-connected cubic graph G and vertex r, can the copies of $2G - \delta(r)$ be partitioned into spanning trees T_1, T_2, T_3 so that, after permuting coordinates,

$$|\delta(Q) \cap K(T_1) \cap K(T_2)| \equiv 0 \pmod 2$$

for every component Q of $K(T_3)$?

Theorem 3.4 supplies the tree partition. Corollary 5.2 says that the additional parity condition is exactly a five-point compatible labelling in the canonical coordinate plane. Proposition 3.5 allows navigation throughout the fixed fibre. What is missing is a proof that one state satisfies the nonlinear quotient-boundary condition.

Theorem 6.8 supplies a genuine reduction for this missing statement: outside the prescribed root, triangles may be contracted without changing whether the fibre has a good state. Thus a prescribed-root minimal simple 3-edge-connected obstruction has no triangle avoiding the root, and an unrooted minimal obstruction may be taken triangle-free. This reduction does not select the three cographic bases; it only removes one local configuration while preserving the exact target.

The parity-element and cographic-base forms expose additional structure, but Theorem 6.5 shows that their most direct graphic-closure strengthening is still too strong. The surviving exchange route is the following.

Open problem 9.2 (universal symmetric descent)**.** Does Theorem 8.1 hold without the order bound?

A positive answer would also prove the star-fibre selection statement: starting from any feasible packing, repeated equal-level navigation and strict descent reaches $d_{\min} = 0$. The finite census supplies no minimal-counterexample reduction and therefore does not answer this problem.

The evidence and no-go results sharply delimit plausible approaches:

1. One cannot force $K(T_i) \cap K(T_j) = \varnothing$, by Proposition 6.1.

2. One cannot choose an arbitrary T_3 and repair only the other two trees, by Proposition 6.2.

3. One cannot independently choose parity-good perfect forests and invoke an unconditional simultaneous extension theorem, by Proposition 6.3.

4. One cannot use the terminal condition that some Fano direction has size at most one, by Theorem 7.1.

5. One cannot demand monotone descent for one preselected Fano coordinate; the 14-vertex state in Section 8.4 is a trapped singleton at defect two.

6. One cannot replace quotient evenness by graphic closure, even in a vertex-star fibre, by Theorem 6.5. The smaller Proposition 6.4 shows the same failure for a generic type-A fibre, and Theorem 6.6 gives an infinite rooted family of star failures. Those closure failures do not survive as exact obstructions: Theorem 6.8 proves that exact component parity lifts as well as descends.

7. One cannot require an immediately descending reciprocal exchange: the first order-16 state above has none.

8. One cannot repair immediate descent by lexicographically minimizing the sorted seven-defect profile: the order-36 state above has 0/13/50 lower/equal/higher legal neighbours. Its exact two-step same-level escape leaves the full plateau-component problem open.

9. One cannot repair immediate descent by moving only among realizations of one fixed odd-kernel triple: the second order-16 state has no such neighbour. A successful universal descent argument must exploit the full same-level component, including active neutral exchanges, and must control the simultaneous topology changes in (8.3).

10. One cannot prove a four-cycle reduction by lifting one fixed good downstairs state inside the square, even if the successful coordinate may change, by Proposition 6.9. The whole-fibre existential square implication is still open, though the fixed-cover criterion (6.22) is exact and the stronger local-selection statement has no countermodel through downstairs order fourteen.

The current computations support Problem 9.1, but they do not exclude a larger countermodel to it. Conversely, failure of this particular star-fibre lemma would not be a counterexample to FiveCDC: another multiplicity fibre or a non-Jaeger cover could still succeed.

10 Reproducibility and certificate boundary

All paths below are relative to the workspace root containing this preprint.

Claim	Primary artifact or checker	Verification level
Encoding equivalence	`docs/encoding.md`; fixed-graph Lean development under `formal/FiveCDC`	human + formal
Component criterion	`scratch/jaeger-support5-component-criterion.md`; `scratch/verify_jaeger_support5_component_criterion.py`	human + exhaustive; cf. [6]
Odd-kernel identities and Petersen no-go	`scratch/jaeger-k-forest-star-parity-identities.md`; `scratch/verify_jaeger_k_forest_petersen.py`	human + exhaustive
Cube fixed-base no-go	`scratch/jaeger-fixed-third-tree-parity-no-go.md`; `scratch/check_jaeger_fixed_third_tree_parity_no_go.py`	human + exhaustive
Perfect-forest extension no-go	`scratch/jaeger-perfect-forests-first-no-go.md`; `scratch/check_jaeger_perfect_forests_first_no_go.py`	human + exhaustive
Parity-element and cographic-base reformulations	`scratch/jaeger-parity-element-and-kernel-closure-no-go.md`	human proof
Fixed-coordinate descent no-go	`scratch/jaeger-star-parity-descent-countermodel.md`; `scratch/verify_jaeger_star_parity_descent_countermodel.py`	literal exhaustive
Symmetric descent through order 14	`output/jaeger-fano-min-descent-through14/census.jsonl`; `scratch/verify_jaeger_fano_min_descent_frontier.py`	whole-state + coverage
Targeted order-16 symmetric descent	`output/jaeger-fano-min-descent-order16-closure-no-go/`; `scratch/verify_jaeger_fano_min_descent_order16_closure_no_go.py`	whole-state + local replay

Claim	Primary artifact or checker	Verification level
Seven-plane exchange law and two local no-gos	`scratch/jaeger-fano-reciprocal-exchange-law.md`; `scratch/verify_jaeger_fano_min_immediate_descent_countermodel.py`	human + literal exhaustive
Sorted-profile one-step no-go	`scratch/jaeger-sorted-profile-local-no-go-order36.md`; `scratch/verify_jaeger_sorted_profile_local_no_go_order36.py`	literal exhaustive
Triangle-expanded descent controls	`output/jaeger-fano-min-descent-triangle-expansions-16v/`; `scratch/verify_jaeger_triangle_expanded_fixedtrap_descent.py`	whole-state + coverage
Generic type-A closure no-go	`scratch/verify_jaeger_kernel_closure_typea_countermodel.py`	independent exhaustive
Vertex-star closure no-go	`output/jaeger-star-kernel-closure-countermodel-16v/`; `scratch/verify_jaeger_star_kernel_closure_countermodel_16v.py`; `scratch/count_jaeger_star_kernel_closure_countermodel_16v.cpp`	CNF + LRAT + exhaustive
Triangle-expansion closure family	`scratch/jaeger-star-kernel-closure-triangle-expansion.md`; `scratch/verify_jaeger_star_kernel_closure_triangle_expansion.py`	human + literal replay
Exact triangle invariance	`scratch/jaeger-star-exact-parity-triangle-invariance.md`; `scratch/verify_jaeger_star_exact_parity_triangle_lift.py`	human + exhaustive
Square-local statewise lifting no-go	`scratch/jaeger-star-square-any-coordinate-lift-no-go.md`; `scratch/verify_jaeger_star_square_any_coordinate_lift_countermodel.py`	human + exhaustive
Square-lift finite boundary	`scratch/verify_jaeger_star_square_order6_controls.py`; `scratch/verify_jaeger_star_square_existential_order8.py`; `scratch/search_jaeger_star_square_existential_census.py`; `scratch/verify_jaeger_star_square_existential_order14.py`	human fixed-cover lemma + certificate census through order 14
Simultaneous triangle product	`scratch/jaeger-simultaneous-triangle-lift-plateau-search.md`; `scratch/search_jaeger_lifted_immediate_traps.py`	human + finite search
34-vertex thinning no-go	`output/jaeger-star-thinning-countermodel-34v/star-thin-any.cnf` and `star-thin-any.lrat`; `scratch/verify_jaeger_star_thinning_countermodel_34v.py`	CNF + dual LRAT

Claim	Primary artifact or checker	Verification level
34-vertex positive five support	`star-good-five-witness.json`; `scratch/verify_jaeger_34v_star_good_witness.py`	literal replay
All fibres through order 14	`output/jaeger-five-point-fibres-through14/census.jsonl`; `scratch/verify_jaeger_five_point_frontier.py`	rerunnable aggregate
Order-38 star census	`scratch/jaeger-coordinate-five-star-order38-result.json`	producer checked
Order-44 selected family	`output/jaeger-coordinate-five-order44/witnesses.jsonl`; `scratch/verify_jaeger_coordinate_five_order44.py`	literal replay

The short commands are collected in `preprint-jaeger-fivecdc-frontier/REPRODUCIBILITY.md`. The positive order-14 and order-38 results have no UNSAT claim and no proof trace; their trust boundary is the generator, SAT solver, internal model checks, and aggregate verifier. The order-44 archive is stronger because its models are retained and reconstructed by independent code. The 34-vertex negative result is stronger still in a different direction: its complete UNSAT formula is accompanied by a proof trace checked by two independently implemented checkers, one formally verified. The symmetric-descent census has an independently checked canonical coverage manifest, but only one implementation enumerates its half-billion exact states. The kernel- closure no-go on the generic 12-vertex fibre is exhaustively replayed by a standard-library checker but has no proof trace; the 16-vertex vertex-star no-go has a complete CNF and an LRAT accepted by both certificate checkers. The infinite closure family then follows from the human triangle-contraction proof, not from extrapolating the finite census. The exact triangle-invariance theorem is a separate human reduction whose seven-row local table is exhaustively replayed by its checker.

11 AI-use disclosure and authorship

This project was conducted with extensive use of OpenAI Codex agents under Atharva Vaidya's direction. The agents:

- translated the research prompt into exact SAT/XOR and graph-theoretic predicates;
- proposed the fixed-fibre, component-parity, odd-kernel, parity-element, cographic-base, exchange, symmetric-descent, and perfect-forest formulations, and the exact triangle-lifting reduction, square-extension criterion, and square-local no-go tests;
- wrote most of the search, certificate-generation, and verification code;
- found the finite countermodels and positive witnesses;
- ran literature searches and assembled bibliographic metadata; and
- drafted essentially all prose and LaTeX in this manuscript.

The human-directed protocol required explicit conjecture/proof/certificate labels, independent semantic checkers, and prominent nonresolution warnings. Nevertheless, AI generation is not independent mathematical review. No claim is made that a human author independently derived every argument before this draft was prepared. A specialist should verify the proofs, software semantics, certificate toolchain, and prior-art boundary before submission. AI systems are not listed as authors; responsibility for any public version rests with the human author.

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